

Laboratoriya 14

Mavzu: Formani JavaScript orqali tekshirish (validation)

Ishning maqsadi:

Bu laboratoriya ishining maqsadi — veb-formalarni JavaScript yordamida mijoz tomonida (client-side) tekshirish usullarini o'rganish, ular orqali foydalanuvchi kiritgan ma'lumotlarni format, bo'sh yoki noto'g'ri qiymatlardan himoya qilish, hamda foydalanuvchi uchun qulay va xavfsiz interfeys yaratish. Talaba quyidagilarni egallashi kerak:

HTML forma elementlari va ularning atributlarini (type, required, pattern va h.k.) to'g'ri ishlatishni bilish;

DOM orqali formaga murojaat qilish va hodisalar bilan (events) ishlashni o'rganish;

Constraint Validation API va setCustomValidity, checkValidity, reportValidity funksiyalaridan foydalanishni tushunish;

Regulyar ifodalar (regex) yordamida qoida belgilash;

Klient va server tomonlama tekshiruvlar farqini tushunish va xavfsizlik talablarini inobatga olish.

Nazariy qism:

Forma nima va uning rollari

Veb-forma — foydalanuvchidan ma'lumot olish uchun mo'ljallangan HTML elementlar to'plami (<form>, <input>, <textarea>, <select> va boshqalar). Formaning asosiy vazifasi — foydalanuvchi kiritgan ma'lumotni serverga yuborish yoki JavaScript orqali qayta ishlash.

Client-side (mijoz tomon) vs Server-side (server tomon) validation

Client-side validation: JavaScript yoki HTML atributlari (required, pattern, type) orqali brauzer darajasida bajariladi. Foydalanuvchi tajribasini yaxshilaydi (tez feedback), tarmoq so'rovlarini kamaytiradi.

Cheklov: foydalanuvchi JavaScriptni o'chirishi yoki so'rovni to'g'rilab yuborishi mumkin — shu sababli server taraf ham tekshirishi shart.

Server-side validation: Ma'lumot serverga yetib borgach, server tilida tekshiriladi (mas., PHP, Node.js). Bu — yakuniy xavfsizlik qatlamidir va dati tozalash (sanitization) ham shu bosqichda amalga oshiriladi.

HTML5 constraint validation (cheklov API)

HTML5 brauzerlar uchun bir qator atributlar taqdim etadi:

required — maydon bo'sh bo'lmasligi kerak.

type — email, number, tel, url kabi qiymat turlarini belgilash.

pattern — regex ko'rinishida kiritish qoidalari.

min, max, minlength, maxlength, step — raqam va uzunlik cheklovlari.

Brauzer shu atributlarga asosan avtomatik tekshiradi va xatolik xabarlarini ko'rsatadi.

Constraint Validation API (JavaScript orqali)

Form va input elementlarining muhim xossalari va metodlari:

`element.checkValidity()` — `element` uchun belgilangan qoidalar bajarilsa `true`, aks holda `false`.

`form.checkValidity()` — butun formani tekshiradi.

`element.reportValidity()` — agar noto'g'ri bo'lsa natijani foydalanuvchiga ko'rsatadi (brauzer xabar oynasi).

`element.validity` — `ValidityState` obyekti: `valid`, `valueMissing`, `typeMismatch`, `patternMismatch`, `tooLong`, `tooShort`, `rangeUnderflow`, `rangeOverflow`, `stepMismatch`, `badInput`, `customError`.

`element.setCustomValidity(message)` — o'z xatolik xabarini o'rnatish (bo'sh string = yo'q xato).

DOM hodisalari (events)

Eng muhimlari: `submit` — forma yuborilganda (formadagi `submit` tugmasi bosilganda yoki `Enter` bosilganda yuz beradi). Bu hodisada `event.preventDefault()` bilan yuborishni to'xtatib, tekshiruvni amalga oshirish mumkin.

`input` — maydon qiymati har o'zgarishda chaqiriladi — real-vaqt feedback uchun foydali.

`change` — elementdan fokus ketganda va qiymat o'zgarganda.

`blur` va `focus` — foydalanuvchi elementga kirdi va chiqdi.

Amaliy qism:

Html kod:

```
<!DOCTYPE html>
<html lang="uz">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Talabalar Ro'yxati</title>
  <!-- Google Fonts -->
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap"
rel="stylesheet">
  <!-- CSS -->
  <link rel="stylesheet" href="style.css">
  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
</head>

<body>

  <div class="container">
```

```

    <header>
      <h1>Talabalar Ro'yxati</h1>
      <p class="subtitle">Qidirish maydonchasiga F.I.Sh., Guruh, Yo'nalish yoki Baho
bo'yicha ma'lumot kiriting:
    </p>
    </header>

    <div class="search-container">
      <input type="text" id="searchInput" placeholder="Qidirish...">
    </div>

    <div class="table-container">
      <table id="studentTable">
        <thead>
          <tr>
            <th>T/R</th>
            <th>Talaba F.I.SH.</th>
            <th>Guruh</th>
            <th>Yo'nalish</th>
            <th>Fan</th>
            <th>Semestr</th>
            <th>O'qituvchi</th>
            <th>Bahosi</th>
          </tr>
        </thead>
        <tbody>
          <!-- Rows will be populated by jQuery -->
        </tbody>
      </table>
      <div id="noResults" class="no-results" style="display: none;">
        Hech qanday ma'lumot topilmadi.
      </div>
    </div>

    <script src="script.js"></script>
  </body>
</html>

```

Script.js kod:

```

$(document).ready(function() {
  // 1. Generate Dummy Data (30 students)
  const students = [];
  const firstNames = ["Asadbek", "Diyorbek", "Shaydaxon", "Ruxshona", "Sevinch",
"Sirojiddin", "Gulshan", "Azizbek", "Madina", "Jasurbek", "Otabek", "Nilufar", "Bobur",
"Malika", "Sardor", "Zilola", "Farrux", "Dilnoza", "Jamshid", "Shahnoza", "Umid", "Kamola",
"Bekzod", "Laylo", "Sherzod", "Nargiza", "Javohir", "Feruza", "Sanjar", "Barno"];
  const lastNames = ["Achilov", "Alimov", "Baltayeva", "Baxtiyorova", "Bekchanova",
"Boltaboyev", "Egamberganova", "Karimov", "Tursunova", "Rahimov", "Abdullayev", "Yusupova",
"Ismailov", "Saidova", "Nazarov", "Qodirova", "Rustamov", "Xolmatova", "Mirzayev", "Ahmedova",

```

```

"Sultonov", "Umarova", "Rasulov", "Aliyeva", "Hasanov", "Ibragimova", "Shukurov", "Mahmudova",
"Olimov", "G'ofurova"];

const middleNames = ["Rajapbayevich", "Qilichboy o'g'li", "Otanazar qizi", "Mansurbek
qizi", "Safarboy qizi", "O'ktam o'g'li", "Otabek qizi", "Alisher o'g'li", "Rustam qizi",
"Bahodir o'g'li", "Ravshan o'g'li", "Dilshod qizi", "Anvar o'g'li", "Farhod qizi", "Ilhom
o'g'li", "Botir qizi", "Ulug'bek o'g'li", "Shuhrat qizi", "Akmal o'g'li", "Jalol qizi",
"Baxtinur o'g'li", "Sobir qizi", "Nodir o'g'li", "Azim qizi", "Davron o'g'li", "Zafar qizi",
"Murod o'g'li", "Hamid qizi", "Temur o'g'li", "Erkin qizi"];

const directions = ["KIDT", "Axborot Xavfsizligi", "Dasturiy Injiniring",
"Telekommunikatsiya"];

const subjects = ["Web Texnologiyalar", "Ma'lumotlar Bazasi", "Algoritmlar",
"Kiberxavfsizlik", "Sun'iy Intellekt"];

const teachers = ["A. Oripov", "B. Qosimov", "D. Eshmatov", "G. Toshmatova", "S.
Rahimov"];

for (let i = 1; i <= 30; i++) {
  const randomFirstName = firstNames[Math.floor(Math.random() * firstNames.length)];
  const randomLastName = lastNames[Math.floor(Math.random() * lastNames.length)];
  const randomMiddleName = middleNames[Math.floor(Math.random() * middleNames.length)];

  students.push({
    id: i,
    fullName: `${randomLastName} ${randomFirstName}
${randomMiddleName}`.toUpperCase(),
    group: "234",
    direction: directions[Math.floor(Math.random() * directions.length)],
    subject: subjects[Math.floor(Math.random() * subjects.length)],
    semester: "5-semester",
    teacher: teachers[Math.floor(Math.random() * teachers.length)],
    grade: Math.floor(Math.random() * 3) + 3 // Random grade between 3 and 5
  });
}

// 2. Render Table
const tableBody = $('#studentTable tbody');
const noResults = $('#noResults');

function renderTable(data) {
  tableBody.empty();

  if (data.length === 0) {
    noResults.show();
  } else {
    noResults.hide();
    data.forEach(student => {
      const row = `
        <tr>
          <td>${student.id}</td>
          <td>${student.fullName}</td>
          <td>${student.group}</td>
          <td>${student.direction}</td>

```

```

        <td>${student.subject}</td>
        <td>${student.semester}</td>
        <td>${student.teacher}</td>
        <td>${student.grade}</td>
    </tr>
    `;
    tableBody.append(row);
  });
}
}

// Initial render
renderTable(students);

// 3. Search Functionality
$('#searchInput').on('keyup', function() {
  const value = $(this).val().toLowerCase();

  const filteredStudents = students.filter(student => {
    // Check if any property value contains the search term
    return Object.values(student).some(val =>
      String(val).toLowerCase().includes(value)
    );
  });

  renderTable(filteredStudents);
});
});

```

Style.css kod:

```

:root {
  --primary-color: #0b6324;
  --primary-hover: #0d3904;
  --bg-color: #f8f9fa;
  --text-color: #333;
  --table-header-bg: #615209;
  --table-header-text: #ffffff;
  --row-hover-bg: #f1f1f1;
  --border-color: #dee2e6;
  --shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
  --radius: 8px;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: 'Inter', sans-serif;

```

```
    background-color: var(--bg-color);
    color: var(--text-color);
    line-height: 1.6;
    padding: 20px;
}
.container {
    max-width: 1200px;
    margin: 0 auto;
    background: #ffffff;
    padding: 30px;
    border-radius: var(--radius);
    box-shadow: var(--shadow);
}
header {
    text-align: center;
    margin-bottom: 30px;
}
h1 {
    color: var(--primary-color);
    font-size: 2rem;
    margin-bottom: 10px;
    font-weight: 700;
}
.subtitle {
    color: #666;
    font-size: 1rem;
}

.search-container {
    margin-bottom: 20px;
}
#searchInput {
    width: 100%;
    padding: 15px;
    font-size: 1rem;
    border: 2px solid #e9ecef;
    border-radius: var(--radius);
    transition: border-color 0.3s ease, box-shadow 0.3s ease;
    outline: none;
    background-color: #f8faff;
}
#searchInput:focus {
    border-color: var(--primary-color);
    box-shadow: 0 0 0 3px rgba(0, 123, 255, 0.1);
    background-color: #fff;
}
.table-container {
    overflow-x: auto;
    border-radius: var(--radius);
    border: 1px solid var(--border-color);
}
```

```

table {
  width: 100%;
  border-collapse: collapse;
  min-width: 800px;
  /* Ensure table doesn't squish too much on small screens */
}
thead {
  background-color: var(--table-header-bg);
  color: var(--table-header-text);
}
th,
td {
  padding: 15px;
  text-align: left;
  border-bottom: 1px solid var(--border-color);
}
th {
  font-weight: 600;
  text-transform: uppercase;
  font-size: 0.85rem;
  letter-spacing: 0.5px;
}
tbody tr {
  transition: background-color 0.2s ease;
}
tbody tr:hover {
  background-color: var(--row-hover-bg);
}
tbody tr:last-child td {
  border-bottom: none;
}
/* Specific column widths if needed */
th:nth-child(1) {
  width: 5%;
}
/* T/R */
th:nth-child(2) {
  width: 25%;
}
/* F.I.SH */
th:nth-child(8) {
  width: 8%;
}
/* Baho */
.no-results {
  text-align: center;
  padding: 20px;
  color: #666;
  font-style: italic;
}
/* Responsive adjustments */

```

```
@media (max-width: 768px) {
  .container {
    padding: 15px;
  }
  h1 {
    font-size: 1.5rem;
  }
  th,
  td {
    padding: 10px;
  }
}
```

Natija:

T/R	TALABA F.I.SH.	GURUH	YO'NALISH	FAN	SEMESTR	O'QITUVCHI	BAHOSI
1.	RUSTAMOV BEKZOD OTABEK QIZI	234	Dasturiy Injiniroing	Algoritmlar	5-semestr	A. Oripov	3
2.	OLIMOV MALIKA BAXTINUR O'G'LI	234	Dasturiy Injiniroing	Ma'lumotlar Bazasi	5-semestr	G. Toshmatova	4
3.	QODIROVA SARDOR AZIM QIZI	234	Axborot Xavfsizligi	Ma'lumotlar Bazasi	5-semestr	B. Qosimov	4
4.	RASULOV RUXSHONA BOTIR QIZI	234	Telekommunikatsiya	Ma'lumotlar Bazasi	5-semestr	D. Eshmatov	5
5.	SHUKUROV NARGIZA AKMAL O'G'LI	234	Dasturiy Injiniroing	Algoritmlar	5-semestr	S. Rahimov	5

T/R	TALABA F.I.SH.	GURUH	YO'NALISH	FAN	SEMESTR	O'QITUVCHI	BAHOSI
14.	YUSUPOVA ZILOLA NODIR O'G'LI	234	Axborot Xavfsizligi	Algoritmlar	5-semestr	B. Qosimov	4

